

Machine Learning Engineering — Executive Diagnostic Brief

Overview

This laboratory implements five machine-learning paradigms: Multiple Linear Regression, Logistic Regression, Decision Trees, Random Forests, and K-Means Clustering.

Exercise	Metric	Actual	Target	Status
Linear Regression	R ²	0.8695	≥ 0.75	PASS
Linear Regression	RMSE	\$4,501.30	$< \$4,800$	PASS
Logistic Regression	ROC-AUC	0.8506	≥ 0.78	PASS
Logistic Regression	Recall	1.0000	≥ 0.70	PASS
Decision Tree	Accuracy	0.8904	≥ 0.85	PASS
Decision Tree	Depth	5	≤ 5	PASS
Random Forest	OOB	0.8274	≥ 0.88	REVIEW
Random Forest	F1	0.4734	≥ 0.65	REVIEW
K-Means	Silhouette	0.1885	≥ 0.42	REVIEW

1. Insurance Cost Prediction

The enhanced linear regression achieved $R^2 = 0.8695$ with $RMSE = \$4,501.30$. Residual diagnostics should be reviewed for increasing variance at higher predicted expense levels. Log-transformed target modeling was also evaluated.

2. Direct Marketing Conversion

Logistic Regression achieved $ROC-AUC = 0.8506$. The classification threshold was calibrated to 0.35, with recall = 1.0000. This supports prioritizing higher-probability prospects while accounting for outreach cost.

3. Decision Tree

The pruned decision tree achieved test accuracy = 0.8904 with depth = 5. The resulting rules provide interpretable decision paths for business users.

4. Random Forest

The tuned Random Forest produced an OOB score of 0.8274 and test F1 = 0.4734. Feature importance and permutation importance were evaluated to identify influential predictors.

5. Wholesale Customer Segmentation

The four-cluster K-Means solution produced a silhouette score of 0.1885. Cluster centroid profiles were calculated using the original euro spending values across six product categories.

Business Recommendations

- Use the regression model as a baseline for insurance-cost forecasting and investigate residual variance before production deployment.
- Use the calibrated marketing classifier to prioritize leads while monitoring precision-recall trade-offs.
- Use the pruned decision tree where interpretability is more important than maximum predictive complexity.
- Use Random Forest feature rankings to support predictor selection and model interpretation.
- Use K-Means centroid profiles to design differentiated inventory and credit-line strategies for customer segments.